

Lausanne *en mutation*

Is the city's housing stock evolving
to match its changing social fabric?

An interactive data story about demographic change, household transformation,
and housing mismatch across Lausanne's districts.

Lens 1
Who lives here

Lens 2
How people live

Lens 3
What housing offers

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<https://lausanne-evolution-team.github.io/lausanne-evolution/website/>

1. Project Overview & Motivation

Lausanne is quietly going through one of the most significant social transformations in its history. Over the past four decades it has absorbed wave after wave of international residents — today nearly half the population is foreign-born, up from 23% in 1979. At the same time, the way people live has changed profoundly: single-person households now account for 49% of all private households, and the average household has shrunk to fewer than two people.

What makes this interesting — and troubling — is what has *not* changed. The housing stock grew by only 12% between 2010 and 2024, and the distribution of apartment sizes barely shifted. A city of singles is being served by a housing stock built for families.

Our central question is: *is Lausanne's housing supply adapting to the way its population is changing?* To answer it, we built an interactive data story structured around three lenses. Lens 1 traces the demographic transformation since 1979. Lens 2 characterises how household composition and employment patterns have changed. Lens 3 examines the housing supply — and reveals the mismatch.

The target audience is twofold: residents who are curious about their city's trajectory, and planners or policymakers who need to understand whether housing production is aligned with demographic demand.

Core finding: In 2024, 49% of Lausanne households are one-person — but only 43% of apartments have 1–2 rooms. This 6-point gap has persisted for over a decade with no sign of closing.

2. Data Sources & Preprocessing

All six datasets come from **Statistique Lausanne**, the city's official statistical office, published as annual Excel tables broken down by statistical district (*quartier*).

Dataset	Description	Years
Population by sex + origin	Swiss / foreign, male / female, by district	1979–2025
Population by age class	Five age groups	1979–2025
Households by size	1 to 6+ persons, mean size	2012–2025
Employment by sex	Headcount and full-time equivalents	2011–2023
Dwellings by rooms	1 to 6+ rooms	2011–2025
Dwellings by surface	7 categories, <40m ² to 160m ² +	2011–2025

The main preprocessing challenge was the inconsistency of Excel layouts across years — three completely different structures depending on the period. We wrote a Python/pandas script (`clean.py`) that detects the layout by inspecting header rows and handles all three cases automatically, producing six clean tidy CSV files ready for D3.

A second issue was district name formatting: older data uses **11-Chailly** while newer data uses **11 - Chailly**. This caused district overlays to appear only from 2015 onward — a bug we fixed by normalising names at match time with a simple regex.

3. Design Decisions & Evolution

From sketches to final product

Our Milestone 2 sketches established the three-lens narrative and identified the gap area chart as the core visualisation. Milestone 2 produced a working prototype with all six MVP charts. Between M2 and M3, three significant design decisions shaped the final product.

The choropleth became a real interactive map. The M2 prototype used a 6×3 colour grid as a placeholder. For M3, we integrated official Lausanne district boundaries from the city’s open data portal. The main hurdle was the coordinate system: Lausanne’s GeoJSON uses EPSG:2056 (Swiss national grid), not WGS84. Using `d3.geoMercator` produced an incorrect rendering; the fix was `d3.geoIdentity().reflectY(true).fitExtent()`, which treats the Swiss coordinates as a flat plane. We also added zoom-in, zoom-out, and reset controls. The map became interactive: clicking any district filters the other charts.

The gap chart became a scrollytelling experience. In M2 it was a standalone card. We redesigned it as a sticky chart with five scrolling narrative steps, each highlighting a different moment: the 2012 baseline, the flat supply response through 2018, the acceleration from 2020, and the full persistent mismatch. Implemented with the Intersection Observer API and D3 transitions.

District-level comparison was added. This was not in the original plan. Once the GeoJSON map was working, we built a global `DistrictFilter` object that propagates a district selection across all charts. The stacked area chart overlays the district’s foreign-population trend; the dual-line chart redraws completely with a rescaled y-axis. The contrast is striking — Sauvabelin has only 26% single-person households while Centre exceeds 58%.

Visual identity

The site uses *Fraunces* (display serif) for headings and *DM Sans* for body text. The three lenses are colour-coded: blue for demographics, green for households, red for housing supply. Cards fade and slide up on scroll. Chart lines draw left-to-right using `stroke-dashoffset` animation. Hero statistics count up from zero on page load.

What we dropped

The radar chart, waffle chart, and bump chart were listed as extras in M2. We chose not to implement them — the scrollytelling and district interactivity were more valuable, and we preferred to do fewer things well.

4. Technical Implementation

The website is a single HTML page with six JavaScript modules. There is no build step — D3.js v7 and Google Fonts load from CDN, everything else is vanilla ES6+ JavaScript.

On load, `main.js` fetches all six CSVs and the GeoJSON in parallel via `Promise.all`, computes derived values, and passes everything to the lens modules. A global `DistrictFilter` object holds the current filter state and notifies registered callbacks when a district is selected or cleared.

Tool	Role
D3.js v7	All charts, scales, axes, transitions, GeoJSON rendering
Intersection Observer API	Scroll-triggered reveals and scrollytelling step activation
d3.geoIdentity().reflectY(true).fitExtent() + d3.zoom	Swiss EPSG:2056 coordinates, map fitting, and zoom controls
Python · pandas · openpyxl	Layout-aware Excel parsing, tidy CSV output
GitHub Pages	Public deployment from master branch

5. Challenges & How We Solved Them

Excel layout heterogeneity

The six raw files used three completely different structures depending on the year. We solved this by detecting the layout programmatically — inspecting which row contains the header and whether years appear side-by-side or stacked — before parsing.

GeoJSON coordinate system

Lausanne’s official GeoJSON uses EPSG:2056 (Swiss national grid), not WGS84. Any standard D3 projection produced a blank canvas. We switched to `d3.geoIdentity().reflectY(true).fitExtent()` which treats the coordinates as a flat plane and corrects the y-axis direction.

Scrollytelling null crash on GitHub Pages

The initial deployment crashed with `getBoundingClientRect` on `null`. The root cause was that `scrolly.js` queried the chart element before the browser had laid it out. Fixed with null guards (`if (!el) return;`) and a width fallback (`W = el.getBoundingClientRect().width || 520`).

Y-axis clipping when filtering districts

When a district’s 1-person household share (e.g. Sauvabelin at 26–30%) fell below the city-level y-axis minimum (44%), the overlay line was invisible. The solution was to redraw the entire chart on district selection, recomputing the y-domain from the union of city and district values.

District name format inconsistency

Pre-2015 data uses 11-Chailly while post-2015 uses 11 - Chailly. Caused overlays to appear only from 2015. Fixed by normalising all names at comparison time with a regex that strips spaces around hyphens.

6. Screenshots of the Final Product

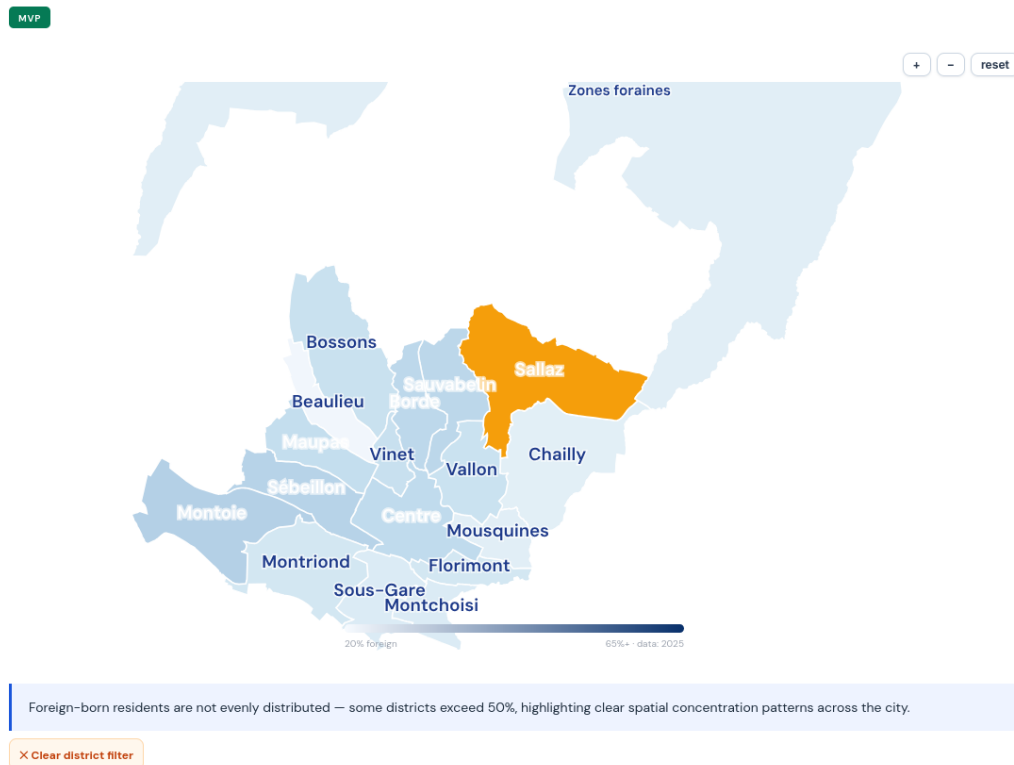
Hero section — key statistics



GeoJSON choropleth map — district selected, zoom controls visible

Foreign resident share by district

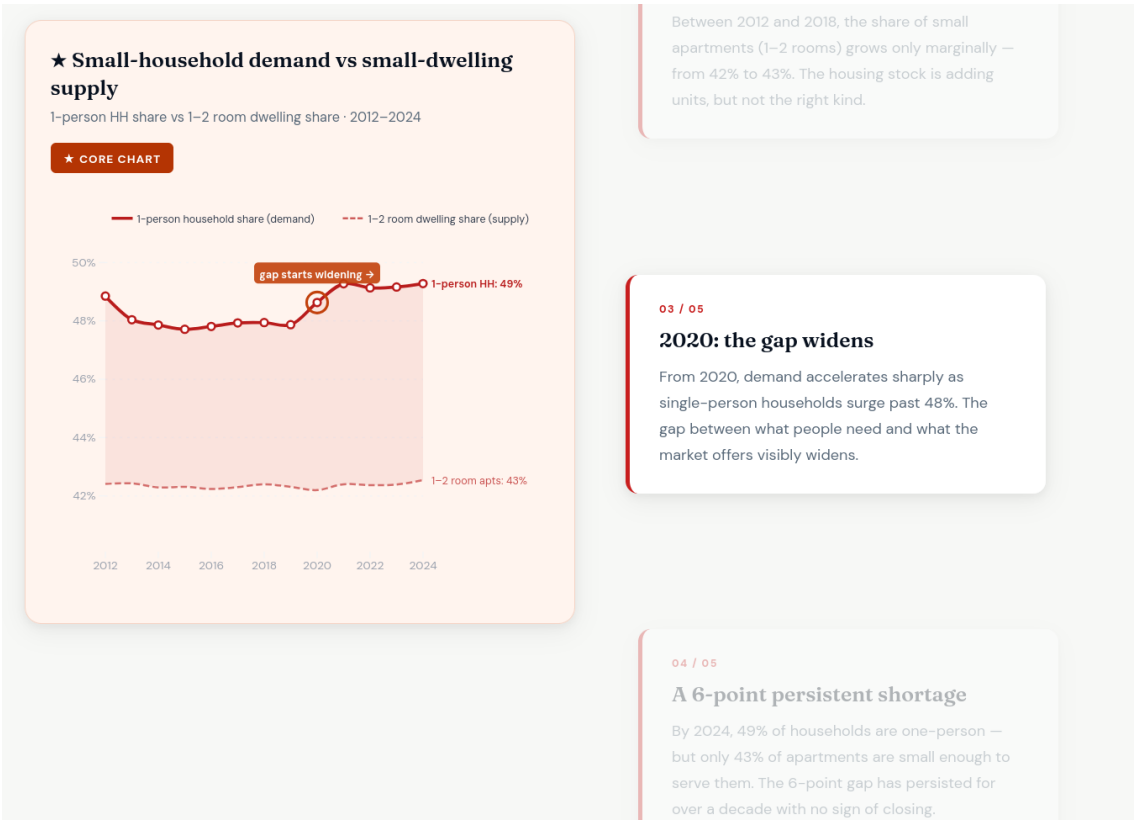
Latest available year · colour intensity = % foreign-born · hover for details



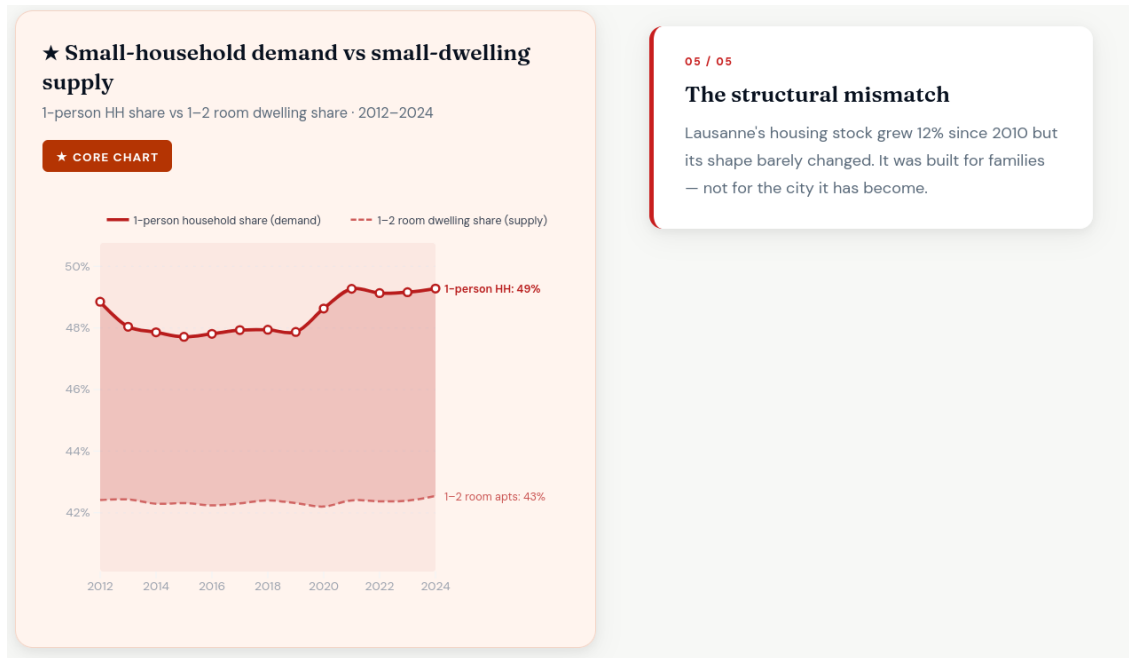
Lens 1 & 2 — Stacked area and single-person households with district overlay



Lens 3 — Scrollytelling step 3: the gap widens (2020)



Lens 3 — Scrollytelling step 5: the structural mismatch



7. Findings & Limitations

The visualisation confirmed and quantified a structural housing mismatch that was hiding in six spreadsheets. The key findings:

- Lausanne's foreign-born share grew from 23% in 1979 to 45% in 2025, with strong spatial concentration — Centre and Sébeillon exceed 55% while Sauvabelin and Boissy stay below 35%.
- Single-person households grew from 42% to 49% between 2012 and 2025 — a 7-point increase in 13 years.
- The share of 1-2 room apartments grew from 42% to 43% over the same period — barely 1 point.
- The resulting 6-point gap has persisted since 2012 and widened noticeably from 2020, with no convergence in sight.
- At district level, Sauvabelin has only 26–30% single-person households while Centre exceeds 58%, reflecting very different housing typologies within the same city.

Limitations. No vacancy data was available at district level, which would have strengthened the mismatch argument. Employment data exists only at city level, preventing district-level filtering in Lens 2. The five age-group categories are fairly broad — finer resolution would better capture student and young professional dynamics.

8. Conclusion

Lausanne en mutation tells a data-driven story that was hiding in plain sight in six spreadsheets. By combining demographic, household, employment, and housing supply data in a single interactive narrative for the first time, the project reveals a structural mismatch that has been

building since at least 2012. The city increasingly lives alone — but its housing stock was built for families.

The scrollytelling sequence, the district-level interactivity, and the GeoJSON map together make the argument accessible to a general audience while remaining rigorous enough for policymakers. The 6-point gap is the project’s core finding: quantified, visualised, and narrated in a way that is impossible to scroll past without understanding.

In one sentence: Lausanne’s social fabric has changed profoundly since 1979 — its housing stock has not.

9. Peer Assessment

For most of the project, the three team members worked together on-site during shared sessions, often using the same laptop. Most commits were pushed from Farah’s account for this reason, so the Git history does not fully reflect the collaborative nature of the work.

Name	Main focus areas	Share
Farah Kacem	Data preprocessing, website integration, GeoJSON map, district interactivity, debugging, and report coordination.	34%
Zyad Aoutir	Demographic and household visualisations (Lens 1 & 2), chart design, CSS layout, and visual refinements.	33%
Anass Inani	Housing-supply visualisations (Lens 3), gap analysis, scrollytelling, screencast production, and documentation.	33%

Most design choices, debugging steps, and integration decisions were made collectively during shared working sessions.

10. How to Run the Project

Local development requires a local HTTP server (browsers block CSV loading from `file://`):

```
python -m http.server 8000
# then open: http://localhost:8000/website/
```

To regenerate the processed CSVs from the raw Excel files:

```
pip install pandas openpyxl
python preprocessing/clean.py
```

Repository: `data/raw/` — original Excel files · `data/processed/` — six clean CSVs + GeoJSON · `preprocessing/clean.py` — preprocessing script · `website/` — HTML, CSS, JS, and data copies for GitHub Pages.

Dependencies: No build step required. D3.js v7 and Google Fonts load from CDN. All other code is vanilla JavaScript ES6+, compatible with all modern browsers.